The free nilpotent group

Terence Tao · 21 December 2009

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In a multiplicative group G, the commutator of two group elements g, h is defined as $[g,h] := g^{-1}h^{-1}gh$ (other conventions are also in use, though they are largely equivalent for the purposes of this discussion). A group is said to be *nilpotent of step s* (or more precisely, step $\leq s$), if all iterated commutators of order $s+1$ or higher necessarily vanish. For instance, a group is nilpotent of order 1 if and only if it is abelian, and it is nilpotent of order 2 if and only if $[[g_1,g_2],g_3] = id$ for all g_1, g_2, g_3 (i.e. all commutator elements $[g_1,g_2]$ are central), and so forth. A good example of an s-step nilpotent group is the group of $s+1 \times s+1$ upper-triangular unipotent matrices (i.e. matrices with 1s on the diagonal and zero below the diagonal), and taking values in some ring (e.g. reals, integers, complex numbers, etc.).

Another important example of nilpotent groups arise from operations on polynomials. For instance, if $V_{\leq s}$ is the vector space of real polynomials of one variable of degree at most s, then there are two natural affine actions on $V_{\leq s}$. Firstly, every polynomial Q in $V_{\leq s}$ gives rise to an "vertical" shift $P \mapsto P+Q$. Secondly, every $h \in \mathbf{R}$ gives rise to a "horizontal" shift $P \mapsto P(\cdot + h)$. The group generated by these two shifts is a nilpotent group of step $\leq s$; this reflects the well-known fact that a polynomial of degree $\leq s$ vanishes once one differentiates more than s times. Because of this link between nilpotentcy and polynomials, one can view nilpotent algebra as a generalisation of polynomial algebra.

Suppose one has a finite number $g_1, \ldots, g_n$ of generators. Using abstract algebra, one can then construct the *free nilpotent group* $\mathcal{F}_{\leq s}(g_1, \ldots, g_n)$ of step $\leq s$, defined as the group generated by the $g_1, \ldots, g_n$ subject to the relations that all commutators of order $s+1$ involving the generators are trivial. This is the universal object in the category of nilpotent groups of step $\leq s$ with n marked elements $g_1, \ldots, g_n$. In other words, given any other $\leq s$-step nilpotent group G' with n marked elements $g'_1, \ldots, g'_n$, there is a unique homomorphism from the free nilpotent group to G' that maps each g_j to g'_j for $1 \leq j \leq n$. In particular, the free nilpotent group is well-defined up to isomorphism in this category.

In many applications, one wants to have a more concrete description of the free nilpotent group, so that one can perform computations more easily (and in particular, be able to tell when two words in the group are equal or not). This is easy for small values of s. For instance, when $s = 1$, $\mathcal{F}_{\leq 1}(g_1, \ldots, g_n)$ is simply the free abelian group generated by $g_1, \ldots, g_n$, and so every element g of $\mathcal{F}_{\leq 1}(g_1, \ldots, g_n)$ can be described uniquely as

$$g = \prod_{j=1}^{n} g_j^{m_j} := g_1^{m_1} \ldots g_n^{m_n} \tag{1}$$

for some integers $m_1, \ldots, m_n$, with the obvious group law. Indeed, to obtain existence of this representation, one starts with any representation of g in terms of the generators $g_1, \ldots, g_n$, and then uses the abelian property to push the g_1 factors to the far left, followed by the g_2 factors, and so forth. To show uniqueness, we observe that the group G of formal abelian products $\{g_1^{m_1} \ldots g_n^{m_n} : m_1, \ldots, m_n \in \mathbf{Z}\} \equiv \mathbf{Z}^k$ is already a ≤ 1-step nilpotent group with marked elements $g_1, \ldots, g_n$, and so there must be a homomorphism from the free group to G. Since G distinguishes all the products $g_1^{m_1} \ldots g_n^{m_n}$ from each other, the free group must also.

It is only slightly more tricky to describe the free nilpotent group $\mathcal{F}_{\leq 2}(g_1, \ldots, g_n)$ of step ≤ 2. Using the identities

$$gh = hg[g,h]; \quad gh^{-1} = ([g,h]^{-1})^{g^{-1}} h^{-1} g; \quad g^{-1}h = h[g,h]^{-1}g^{-1}; \quad g^{-1}h^{-1} := [g,h]g^{-1}h^{-1}$$

(where $g^h := h^{-1}gh$ is the conjugate of g by h) we see that whenever $1 \leq i < j \leq n$, one can push a positive or negative power of g_i past a positive or negative power of g_j, at the cost of creating a positive or negative power of $[g_i, g_j]$, or one of its conjugates. Meanwhile, in a ≤ 2-step nilpotent group, all the commutators are central, and one can pull all the commutators out of a word and collect them as in the abelian case. Doing all this, we see that every element g of $\mathcal{F}_{\leq 2}(g_1, ..., g_n)$ has a representation of the form

$$g = (\prod_{j=1}^{n} g_j^{m_j})(\prod_{1 \leq i < j \leq n} [g_i, g_j]^{m_{[i,j]}}) \tag{2}$$

for some integers m_j for $1 \leq j \leq n$ and $m_{[i,j]}$ for $1 \leq i < j \leq n$. Note that we don't need to consider commutators $[g_i, g_j]$ for $i \geq j$, since

$$[g_i, g_i] = id$$

and

$$[g_i, g_j] = [g_j, g_i]^{-1}.$$

It is possible to show also that this representation is unique, by repeating the previous argument, i.e. by showing that the set of formal products

$$G := \{(\prod_{j=1}^{k} g_j^{m_j})(\prod_{1 \leq i < j \leq n} [g_i, g_j]^{m_{[i,j]}}) : m_j, m_{[i,j]} \in \mathbf{Z}\}$$

forms a ≤ 2-step nilpotent group, after using the above rules to define the group operations. This can be done, but verifying the group axioms (particularly the associative law) for G is unpleasantly tedious.

Once one sees this, one rapidly loses an appetite for trying to obtain a similar explicit description for free nilpotent groups for higher step, especially once one starts seeing that higher commutators obey some non-obvious identities such as the *Hall-Witt identity*

$$[[g, h^{-1}], k]^h \cdot [[h, k^{-1}], g]^k \cdot [[k, g^{-1}], h]^g = 1 \tag{3}$$

(a nonlinear version of the Jacobi identity in the theory of Lie algebras), which make one less certain as to the existence or uniqueness of various proposed generalisations of the representations (1) or (2). For instance, in the free ≤ 3-step nilpotent group, it turns out that for representations of the form

$$g = (\prod_{j=1}^{n} g_j^{m_j})(\prod_{1 \leq i < j \leq n} [g_i, g_j]^{m_{[i,j]}})(\prod_{1 \leq i < j < k \leq n} [[g_i, g_j], g_k]^{n_{[[i,j],k]}})$$

one has uniqueness but not existence (e.g. even in the simplest case $n = 3$, there is no place in this representation for, say, $[[g_1, g_3], g_2]$ or $[[g_1, g_2], g_2]$), but if one tries to insert more triple commutators into the representation to make up for this, one has to be careful not to lose uniqueness due to identities such as (3). One can paste these in by *ad hoc* means in the $s = 3$ case, but the $s = 4$ case looks more fearsome still, especially now that the quadruple commutators split into several distinct-looking species such as $[[g_i, g_j], [g_k, g_l]]$ and $[[[g_i, g_j], g_k], g_l]$ which are nevertheless still related to each other by identities such as (3). While one can eventually disentangle this mess for any fixed n and s by a finite amount of combinatorial computation, it is not immediately obvious how to give an explicit description of $\mathcal{F}_{\leq s}(g_1, ..., g_n)$ uniformly in n and s.

Nevertheless, it turns out that one can give a reasonably tractable description of this group if one takes a polycyclic perspective rather than a nilpotent one – i.e. one views the free nilpotent group as a tower of group extensions of the trivial group by the cyclic group $\mathbf{Z}$. This seems to be a fairly standard observation in group theory – I found it in this book of Magnus, Karrass, and Solitar, via this paper of Leibman – but seems not to be so widely known outside of that field, so I wanted to record it here.

— 1. Generalisation —

The first step is to generalise the concept of a free nilpotent group to one where the generators have different "degrees". Define a *graded sequence* to be a finite ordered sequence $(g_\alpha)_{\alpha \in A}$ of formal group elements g_α, indexed by a finite, totally ordered set A, where each g_α is assigned a positive integer $\deg(g_\alpha)$, which we call the *degree* of g_α. We then define the degree of any formal iterated commutator of the g_α by declaring the degree of $[g,h]$ to be the sum of the degrees of g and h. Thus for instance $[[g_{\alpha_1}, g_{\alpha_2}], g_{\alpha_3}]$ has degree $\deg(g_{\alpha_1}) + \deg(g_{\alpha_2}) + \deg(g_{\alpha_3})$. (The ordering on A is not presently important, but will become useful for the polycyclic representation; note that such ordering has already appeared implicitly in (1) and (2).)

Define the *free $\leq s$-step nilpotent group* $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha \in A})$ generated by a graded sequence $(g_\alpha)_{\alpha \in A}$ to be the group generated by the g_α, subject to the constraint that any iterated commutator of the g_α of degree greater than s is trivial. Thus the free group $\mathcal{F}_{\leq s}(g_1, \ldots, g_k)$ corresponds to the case when all the g_i are assigned a degree of 1.

Note that any element of a graded sequence of degree greater than s is automatically trivial (we view it as a 0-fold commutator of itself) and so can be automatically discarded from that sequence.

We will recursively define the free $\leq s$-step nilpotent group of some graded sequence $(g_\alpha)_{\alpha \in A}$ in terms of simpler sequences, which have fewer low-degree terms at the expense of introducing higher-degree terms, though as mentioned earlier there is no need to introduce terms of degree larger than s. Eventually this process exhausts the sequence, and at that point the free nilpotent group will be completely described.

— 2. Shift —

It is convenient to introduce the iterated commutators $[g, mh]$ for $m = 0, 1, 2, \ldots$ by declaring $[g, 0h] := g$ and $[g, (m+1)h] := [[g, mh], h]$, thus for instance $[g, 3h] = [[[g,h],h],h]$.

> **Definition 1 (Shift)** Let $s \geq 1$ be an integer, let $(g_\alpha)_{\alpha \in A}$ be a non-empty graded sequence, and let α_0 be the minimal element of A. We define the (degree $\leq s$) *shift* $(g_\alpha)_{\alpha \in A'}$ of $(g_\alpha)_{\alpha \in A}$ by defining A' to be formed from A by removing α_0, and then adding at the end of A all commutators $[\beta, m\alpha_0]$ of degree at most s, where $\beta \in A \setminus \{\alpha_0\}$ and $m \geq 1$. For sake of concreteness we order these commutators lexicographically, so that $[\beta, m\alpha_0] \geq [\beta', m'\alpha_0]$ if $\beta > \beta'$, or if $\beta = \beta'$ and $m > m'$. (These commutators are also considered to be larger than any element of $A \setminus \{\alpha_0\}$). We give each $[\beta, m\alpha_0]$ a degree of $\deg(\beta) + m \deg(\alpha_0)$, and define the group element $g_{[\beta, m\alpha_0]}$ to be $[g_\beta, m g_{\alpha_0}]$.

> **Example 1** If $s \leq 3$, and the graded sequence g_a, g_b, g_c consists entirely of elements of degree 1, then the shift of this sequence is given by
>
> $$g_b, g_c, g_{[b,a]}, g_{[b,2a]}, g_{[c,a]}, g_{[c,2a]}$$
>
> where $[b,a], [c,a]$ have degree 2, and $[b,2a]$, $[c,2a]$ have degree 3, and $g_{[b,a]} = [g_b, g_a]$, $g_{[b,2a]} = [g_b, 2g_a]$, etc.

The key lemma is then

> **Lemma 2 (Recursive description of free group)** Let $s \geq 1$ be an integer, let $(g_\alpha)_{\alpha \in A}$ be a non-empty graded sequence, and let α_0 be the minimal element of A. Let $(g_\alpha)_{\alpha \in A'}$ be the shift of of $(g_\alpha)_{\alpha \in A}$. Then $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha \in A})$ is generated by g_{α_0} and $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha \in A'})$, and furthermore the latter group is a normal subgroup of $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha \in A})$ that does not contain g_{α_0}. In other words, we have a semi-direct product representation
>
> $$\mathcal{F}_{\leq s}((g_\alpha)_{\alpha \in A}) = \mathbf{Z} \ltimes \mathcal{F}_{\leq s}((g_\alpha)_{\alpha \in A'})$$

with g_{α_0} being identified with $(1, id)$ and the action of $\mathbf{Z}$ being given by the conjugation action of g_{α_0}. In particular, every element g in $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A})$ can be uniquely expressed as $g = g_\alpha^{n_\alpha} g'$, where $g' \in \mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A'})$.

Proof: The argument will be motivated by the identities

$$g_{\alpha_0}^{-1}[g_\beta, mg_{\alpha_0}]g_{\alpha_0} = [g_\beta, mg_{\alpha_0}][g_\beta, (m+1)g_{\alpha_0}] \tag{4}$$

and its inverse

$$g_{\alpha_0}[g_\beta, mg_{\alpha_0}]g_{\alpha_0}^{-1} = [g_\beta, mg_{\alpha_0}][g_\beta, (m+2)g_{\alpha_0}]\ldots([g_\beta, (m+1)g_{\alpha_0}][g_\beta, (m+3)g_{\alpha_0}]\ldots)^{-1} \tag{5}$$

(note that the products terminate in finite time due to nilpotency). In particular, we of course have

$$[[g_\beta, mg_{\alpha_0}], g_{\alpha_0}] = [g_\beta, (m+1)g_{\alpha_0}]. \tag{6}$$

We now form a semi-direct product

$$G := \mathbf{Z} \ltimes \mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A'}).$$

To do this, we need a conjugation action of $(1, id)$ on $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A'})$. Inspired by the identities (4), (5), we define this action on generators by mapping

$$g_{[\beta,m\alpha_0]} \mapsto g_{[\beta,m\alpha_0]}g_{[\beta,(m+1)\alpha_0]} \tag{7}$$

and inverted via the mapping

$$g_{[\beta,m\alpha_0]} \mapsto (g_{[\beta,m\alpha_0]}g_{[\beta,(m+2)\alpha_0]}\ldots)(g_{[\beta,(m+1)\alpha_0]}g_{[\beta,(m+3)\alpha_0]}\ldots)^{-1}. \tag{8}$$

Note that this map and its inverse preserve the degree of the generators, and so by the definition of the weighted free group this defines an automorphism on $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A'})$. By construction, we have the counterpart of (6):

$$[g_{\beta,m\alpha_0}, (1, id)] = g_{\beta,(m+1)\alpha_0}. \tag{9}$$

The group G is generated by $(1, id)$ together with the g_α for $\alpha \in A'$. Giving $(1, id)$ the degree of $\deg(\alpha_0)$, one checks by hand (using plenty of commutator identities, as well as (9)) that any iterated commutator of generators with total degree greater than s in G is trivial. Thus there is a homomorphism from $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A})$ to G which maps g_{α_0} to $(1, id)$ and g_α to g_α for all $\alpha \in A \setminus \{\alpha_0\}$. Comparing (6) with (9) with, we see that this homomorphism also maps $[g_\beta, mg_{\alpha_0}]$ to $g_{[\beta,m\alpha_0]}$.

On the other hand, there is an obvious homomorphism from $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A'})$ to $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A})$ that maps $g_{[\beta,m\alpha_0]}$ to $[g_\beta, mg_{\alpha_0}]$, and from the compatibility of (4), (5) with (7), (8) we may extend this homomorphism to a homomorphism from G to $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A})$ which also maps $(1, id)$ to g_{α_0}. By checking this homomorphism on generators, we see that this map inverts the previous homomorphism from $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A})$ to G, and so the two groups are isomorphic, as required. □

We can now iterate this. Observe that every time one shifts a non-empty graded sequence, one removes one element (the minimal element g_{α_0}) but replaces it with zero or more elements of higher degree. Iterating this process, we eventually run out of elements of degree one, then degree two, and so forth, until the sequence becomes completely empty. We glue together all the elements encountered this way and refer to the full sequence as the *completion* $(g_\alpha)_{\alpha\in\overline{A}}$ of the original sequence $(g_\alpha)_{\alpha\in A}$. As a corollary of the above lemma we thus have

Corollary 3 (Explicit description of free nilpotent group) Let $s \geq 1$ be an integer, and let $(g_\alpha)_{\alpha\in A}$ be a graded sequence. Then every element g of $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A})$ can be represented uniquely as

$$\prod_{\alpha\in\overline{A}} g_\alpha^{n_\alpha}$$

where n_α is an integer, and $\overline{A}$ is the completion of A.

Example 2 We continue with the sequence g_a, g_b, g_c from Example 1, with $s = 3$. We already saw that shifting once yielded the sequence

$$g_b, g_c, g_{[b,a]}, g_{[b,2a]}, g_{[c,a]}, g_{[c,2a]}.$$

Another shift gives

$$g_c, g_{[b,a]}, g_{[b,2a]}, g_{[c,a]}, g_{[c,2a]}, g_{[c,b]}, g_{[c,2b]}, g_{[[b,a],b]}, g_{[[c,a],b]},$$

and shifting again gives

$$g_{[b,a]}, g_{[b,2a]}, g_{[c,a]}, g_{[c,2a]}, g_{[c,b]}, g_{[c,2b]}, g_{[[b,a],b]}, g_{[[c,a],b]}, g_{[[b,a],c]},$$

$$g_{[[c,a],c]}, g_{[[c,b],c]}.$$

At this point, all remaining terms in the sequence have degree at least two, and further shifting simply removes the first element without adding any new elements. Thus the completion is

$$g_a, g_b, g_c, g_{[b,a]}, g_{[b,2a]}, g_{[c,a]}, g_{[c,2a]},$$

$$g_{[c,b]}, g_{[c,2b]}, g_{[[b,a],b]}, g_{[[c,a],b]}, g_{[[b,a],c]}, g_{[[c,a],c]}, g_{[[c,b],c]}$$

and every element of $\mathcal{F}_{\leq 3}(g_a, g_b, g_c)$ can be uniquely expressed as

$$g_a^{n_a} g_b^{n_b} g_c^{b_c} [g_b, g_a]^{n_{[b,a]}} [g_b, 2g_a]^{n_{[b,2a]}}$$

$$[g_c, g_a]^{n_{[c,a]}} [g_c, 2g_a]^{n_{[c,2a]}} [g_c, g_b]^{n_{[c,b]}} [g_c, 2g_b]^{n_{[c,2b]}}$$

$$[[g_b, g_a], g_b]^{n_{[[b,a],b]}} [[g_c, g_a], g_b]^{n_{[[c,a],b]}} [[g_b, g_a], g_c]^{n_{[[b,a],c]}}$$

$$[[g_c, g_a], g_c]^{n_{[[c,a],c]}} [[g_c, g_b], g_c]^{n_{[[c,b],c]}}.$$

In a recent paper of Leibman, a related argument was used to expand bracket polynomials (a generalisation of ordinary polynomials in which the integer part operation $x \mapsto \lfloor x \rfloor$ is introduced) of degree $\leq s$ in several variables $(x_\alpha)_{\alpha \in A}$ into a canonical basis $(x_\alpha)_{\alpha \in \overline{A}}$, where $\overline{A}$ is the same completion of A that was encountered here. This was used to show a close connection between such bracket polynomials and nilpotent groups (or more precisely, nilsequences).

Comments (28)



Dylan Thurston · 22 December 2009

In Example 2, aren't you missing $g_{[c,2b]}$?
[Oops! Corrected, thanks – T.]



Anonymous · 22 December 2009

I'm probably just confused, but I would've guessed there would be 3+3+8 terms in the completion.



Sean Eberhard · 13 May 2013

Is $g_{[[c,b],c]}$ also missing?
[Corrected, thanks – T.]



Anonymous · 22 December 2009

In a 2-step nilpotent groups, when one pushes g_j pass g_i, one may need to use one of the conjugates of the commutator [g_i,g_j]. But in the representation (2), there are only integers powers of the commutators and (non-trivial) conjugates of them. I don't see why. Is it something obvious that I don't see?

 **Terence Tao** · 22 December 2009

In a 2-step nilpotent group, all commutators are central, and are thus their own conjugates.

 **Anonymous** · 22 December 2009

Oops, I meant no nontrivial conjugates of the commutators in the above post.
By the way, is there anyway that one can edit one's own posts (like the reviews at Amazon.com)?

 **Anonymous** · 25 December 2009

Merry Christmas, Terence Tao!

 **Anonymous** · 25 December 2009

A nilpotent group of step s is a very neat concept!
Do we know the nilpotent subgroup of step 2 of GL(n,\mathbb{K})? I could find them for n=2,3, but after that it seems more difficult.

 **Anonymous** · 3 March 2011

when you define a free nilpotent group, do you not have to make all s step commutators trivial rather than just the commutators involving the generators?

 **Terence Tao** · 3 March 2011

It turns out that either approach gives an equivalent group. (Once one makes all s-step commutators trivial, one can show inductively that the ith group G_i in the lower central series is generated by all the iterated commutators of generators of step i or greater, and in particular that G_{s+1} vanishes.)

 **254A, addendum: Some notes on nilprogressions « What's new** · 17 March 2012

[...] also be profitably viewed as a special type of polycylic group; this is the perspective taken in this previous blog post. Last, but not least, one can view nilpotent groups from a combinatorial group theory perspective, [...]

 **Alexey** · 31 May 2012

Thank you for the article. Not completely clear to me that $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A'})$ is isomorphic to a subgroup of $\mathcal{F}_{\leq s}((g_\alpha)_{\alpha\in A})$, i only see that it maps onto the subgroup. I'll need to think more to understand.

 **Terence Tao** · 31 May 2012

~~From the universality properties, there is a map from the latter group to the former which is the identity on the g_α with $\alpha \in A'$ but sends the g_α with $\alpha \in A \backslash A'$ to the identity. By using the universality properties again one can show that this map is a right inverse of the obvious embedding of the former subgroup into the latter, which gives the injectivity.~~
[Argument incorrect, see comments below – T.]

 **Alexey** · 31 May 2012

This particular argument false: if $s=2$, $A = (a, b)$, $A' = (b, [b,a])$, then if a is mapped to 1, so is $[a,b]$. The group ${\mathcal F}_{\le2}(A')$ is indeed isomorphic to a subgroup of ${\mathcal F}_{\le2}(A)$ and isomorphic to $\mathbb Z^2$, but it is not a retract of ${\mathcal F}_{\le2}(A)$.

 **Terence Tao** · 1 June 2012

Ah, you're right, of course. One has to rearrange the argument by constructing the semi-direct product by abstract nonsense first, rather than by embedding first into the free nilpotent group,

and only establishing the isomorphism between the two groups afterwards. I've rewritten the argument to reflect this.



Alexey · 1 June 2012

Thanks! I will read the new version.



Alexey · 4 June 2012

I do not find formula (5), i find instead: $g_{\alpha_0}[g_\beta, mg_{\alpha_0}]g_{\alpha_0}^{-1} = [g_\beta, mg_{\alpha_0}][g_\beta, (m+2)g_{\alpha_0}]\cdots[g_\beta, (m+1)g_{\alpha_0}]^{-1}$
[Corrected, thanks – T.]

Martin Seysen · 14 December 2013

A description of free nilpotent groups of step s is also given in section 11 (Basic commutators) of the book 'The Theory of Groups' by Marshall Hall, Macmillan Company, 12th printing, 1973. Equivalence of that description with your description is not obvious to me.
Btw., do you know if a presentation of such a group in terms of commutators of shape [...[[g_1,g_2], g_3], ..., g_n] exists? (Of course, some of the g_i may be the same).

Alexander Shamov · 11 November 2014

> $g_{[\beta,m\alpha_0]} \mapsto g_{[\beta,m\alpha_0]}g_{[\beta,(m+1)\alpha_0]}$
> Note that this map and its inverse preserve the degree of the generators
I'm not sure I understand what this means. The map that you described doesn't even take generators to generators...

Alexander Shamov · 11 November 2014

It seems that the key lemma here is $[M_r, M_{r'}] \subset M_{r+r'}$, where M_r is the submonoid generated by the generators of degree $\geq r$, and this can be checked by hand.



DS · 26 November 2018

Alexander Shamov you are right, I had the same problem in that place. This is the heart of the proof so it's a shame the argument is not expanded. A key fact useful here is $[xy,z]=y^{-1}x^{-1} x^z y^z=[x,z]^y[y,z]$
In short, "commutator are bilinear up to conjugation".
Thus when applying the homomorphism defined above to a relation, which is iterated commutators of generators, you get iterated commutators of \emph{products} of generators, but by bilinearity it's a product of conjugates of commutators of generators, which are themselves relations, so the map is well-defined.



Anonymous · 23 August 2025

When does a graded sequence of generators yield a filtration on the group in the natural way? I mean, given the behavior of the degree relative to commutators, it seems natural to expect that the set of generators of degree i is a subgroup G_i for each i, in such a way that $(G_i)_i$ is a filtration on the free graded nilpotent group. However, I don't see that the notion of degree here implies that G_i is a subgroup. Does this subgroup property follow somehow, or does one need to add a condition on the degree function for this to follow (e.g. deg$(gh^{-1})geq min($deg(g),deg(h))?



Terence Tao · 23 August 2025

One takes G_i to be the group generated by all the generators of degree at least i (but the set of generators themselves will almost certainly not be a group itself).

Anonymous · 24 August 2025

Thank you! And sorry for the badly posed question, clearly the "set of generators of degree

$$i$$

" was not going to be a subgroup typically. The fact that the

$$G_i$$

that you mention form a filtration is not trivial, I guess, it involves something like the lemma that Alexander Shamov mentions above, right? If so, do you know a reference in the literature that proves such a statement? Thank you again.



Terence Tao · 24 August 2025

Generally what one needs is that the commutator of a generator of degree i and a generator of j lies in G_{i+j}, which also implies that the conjugate of a generator of degree i by a generator of degree j lies in G_i. From this it is straightforward to see that the G_i is normalized by G_j; then it becomes easy to show that $[G_i, G_j] \leq G_{i+j}$ for all $i, j \geq 1$, since modulo the normal subgroup G_{i+j}, every generator of G_i commutes with every generator of G_j.

Anonymous · 25 August 2025

Something seems counterintuitive about this filtration: suppose the grading we chose does not assign degree i to any generator (but there are generators of degree less than i and also generators of degree greater than i). Then, for this grading, in the filtration you defined we would have $G_i = G_{i+1}$. On the other hand, any iterated degree-exactly-i commutator of generators should fall in G_i, by the property you mention, but typically not every degree-exactly-i commutator has to belong to G_{i+1}, right? If so then we would get the contradiction that G_i strictly includes G_{i+1}... I guess seeing details of the proof of the property "the commutator of a generator of degree i and a generator of lies in G_{i+j}" will clarify what happens here.

Anonymous · 26 August 2025

There seems to be a problem here somewhere, sorry in advance if it's me being confused, but I would like to ask again to make sure. You wrote above that a filtration is obtained by taking G_i to be the group generated by all the generators of degree at least i. But if we choose to give all the generators just degree 1 (as we may), then we would just get $G_1 = G$ and then $G_i = 1$ for $i \geq 2$; then, if this satisfied the filtration property $[G_i, G_j] \leq G_{i+j}$ that you stated in your last reply above, that would imply that $[G, G]$ is trivial and so G would be abelian, which is typically false. Could it be that you meant that the group G_i should be defined as the subgroup generated by iterated commutators of degree at least i in the generators, and the inverses of these commutators? (This larger generating set would include the generators of degree at least i as in your definition of G_i, but also other commutators of degree at least i).



Terence Tao · 26 August 2025

As I said before, one needs to add the axiom that the commutator of a generator of degree i and a generator of degree j can be expressed in terms of generators of degree at least $i + j$. One way to do this is indeed to add in all iterated commutators of the initial generator set into the set of generators, assigning them appropriate degrees.